Solving Linear Equations

Name _____ Date _____ No calculator Version 1

KEY WORDS equation · solve · solution · unknown · inverse operation · expand · like terms

A One and two steps ●○○ ANSWER ON THE SHEET

Remember: Do the same thing to both sides. Undo adding and subtracting before multiplying and dividing.

1. Solve each equation.

a) $x + 7 = 12 \quad x =$ _____ **b)** $x - 4 = 9 \quad x =$ _____ **c)** $3x = 21 \quad x =$ _____

d) $\frac{x}{4} = 5 \quad x =$ _____ **e)** $2x + 3 = 11 \quad x =$ _____ **f)** $5x - 4 = 21 \quad x =$ _____

g) $3x + 7 = 1 \quad x =$ _____ **h)** $4x - 9 = 15 \quad x =$ _____ **i)** $\frac{x}{3} + 2 = 6 \quad x =$ _____

j) $15 - 2x = 7 \quad x =$ _____ **k)** $6x + 5 = -13 \quad x =$ _____ **l)** $\frac{x}{5} - 1 = 3 \quad x =$ _____

B Equations with brackets ●●○ WORK IN YOUR BOOK

Remember: Expand the bracket first, then solve as before.

2. Solve.

a) $3(x + 2) = 21$ **b)** $2(x - 5) = 8$ **c)** $4(2x + 1) = 20$

d) $5(x + 3) = 10$ **e)** $2(3x - 4) = 16$ **f)** $6(x - 2) = 30$

g) $3(2x + 5) = 3$ **h)** $-2(x + 4) = -14$ **i)** $4(3x - 2) = 28$

C The unknown on both sides ●●○ WORK IN YOUR BOOK

Remember: Collect the x terms on one side and the numbers on the other.

3. Solve.

a) $5x + 3 = 2x + 18$ **b)** $7x - 4 = 3x + 12$ **c)** $6x + 1 = 4x - 5$

d) $9x - 5 = 2x + 9$ **e)** $2x + 9 = 5x$ **f)** $3x + 14 = 8x + 4$

g) $4x - 7 = -3x + 14$ **h)** $8x + 6 = 5x - 6$ **i)** $4(x + 1) = 2x + 10$

j) $3(x + 4) = 5x + 8$ **k)** $5(x - 2) = 2(x + 4)$ **l)** $2(3x + 1) = 4(x + 2)$

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D Equations with fractions ●●● WORK IN YOUR BOOK

⚠ **Watch out:** Multiply every term on both sides by the number under the fraction, not just the fraction itself.

4. Solve.

a) $\frac{3x}{4} = 9$

b) $\frac{x+5}{2} = 6$

c) $\frac{2x-1}{3} = 5$

d) $\frac{3x+2}{4} = 5$

e) $\frac{5x-3}{2} = 6$

f) $\frac{x}{2} + \frac{x}{3} = 10$

g) $\frac{x}{3} + \frac{x}{4} = 14$

h) $\frac{x+1}{2} = \frac{x+4}{3}$

i) $\frac{x+3}{2} = \frac{x+7}{4}$

E Problems in words ●●● WORK IN YOUR BOOK

5. Write an equation for each problem, then solve it.

a) I think of a number, multiply it by 4 and then subtract 7. The answer is 29. Find the number.

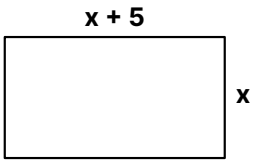
b) Three consecutive whole numbers add to 72. Find the three numbers.

c) A taxi charges \$4 to start, plus \$3 for every kilometre. A trip costs \$31. How far was the trip?

d) Aroha is x years old. Her mother is 3 times as old. Their ages add to 52. How old is each of them?

e) The three angles of a triangle are x°, $2x^\circ$ and $(x+20)^\circ$. Find x.

6. The rectangle has a perimeter of 38 cm. Write an equation and solve it to find x. Then find the length of the rectangle.



★ Extension FOR EARLY FINISHERS

7. Solve $3(2x-1) - 2(x+4) = 9$.

8. Solve $4x + 3 = 10 - 2x$. Give your answer as a fraction.

9. Solve $\frac{x+1}{2} + \frac{x-2}{3} = 4$.

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Solving Linear Equations

Name _____ Date _____ No calculator Version 2

KEY WORDS equation · solve · solution · unknown · inverse operation · expand · like terms

A One and two steps ●○○ ANSWER ON THE SHEET

Remember: Do the same thing to both sides. Undo adding and subtracting before multiplying and dividing.

1. Solve each equation.

a) $x + 9 = 13 \quad x =$ _____ **b)** $x - 6 = 5 \quad x =$ _____ **c)** $4x = 32 \quad x =$ _____

d) $\frac{x}{3} = 6 \quad x =$ _____ **e)** $2x + 5 = 11 \quad x =$ _____ **f)** $4x - 3 = 17 \quad x =$ _____

g) $3x + 10 = 4 \quad x =$ _____ **h)** $5x - 8 = 22 \quad x =$ _____ **i)** $\frac{x}{4} + 3 = 8 \quad x =$ _____

j) $17 - 3x = 5 \quad x =$ _____ **k)** $7x + 4 = -10 \quad x =$ _____ **l)** $\frac{x}{2} - 5 = 4 \quad x =$ _____

B Equations with brackets ●●○ WORK IN YOUR BOOK

Remember: Expand the bracket first, then solve as before.

2. Solve.

a) $2(x + 3) = 18$ **b)** $3(x - 4) = 12$ **c)** $5(2x + 1) = 25$

d) $4(x + 5) = 12$ **e)** $2(4x - 3) = 18$ **f)** $7(x - 2) = 21$

g) $3(2x + 7) = 3$ **h)** $-3(x + 2) = -18$ **i)** $5(2x - 3) = 25$

C The unknown on both sides ●●○ WORK IN YOUR BOOK

Remember: Collect the x terms on one side and the numbers on the other.

3. Solve.

a) $6x + 2 = 3x + 14$ **b)** $8x - 5 = 4x + 7$ **c)** $5x + 3 = 3x - 5$

d) $7x - 6 = 2x + 4$ **e)** $3x + 11 = 7x + 3$ **f)** $2x + 13 = 6x + 1$

g) $5x - 9 = -2x + 12$ **h)** $9x + 8 = 6x - 1$ **i)** $5(x + 1) = 3x + 13$

j) $4(x + 3) = 7x + 6$ **k)** $3(x - 4) = 2(x - 2)$ **l)** $2(5x + 1) = 4(2x + 3)$

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D Equations with fractions ●●● WORK IN YOUR BOOK

⚠ **Watch out:** Multiply every term on both sides by the number under the fraction, not just the fraction itself.

4. Solve.

a) $\frac{2x}{5} = 6$ **b)** $\frac{x+7}{3} = 5$ **c)** $\frac{2x-3}{5} = 3$

d) $\frac{3x+1}{2} = 8$ **e)** $\frac{4x-5}{3} = 5$ **f)** $\frac{x}{2} + \frac{x}{5} = 14$

g) $\frac{x}{3} + \frac{x}{4} = 7$ **h)** $\frac{x+2}{3} = \frac{x+5}{4}$ **i)** $\frac{x+5}{2} = \frac{x+9}{3}$

E Problems in words ●●● WORK IN YOUR BOOK

5. Write an equation for each problem, then solve it.

a) I think of a number, multiply it by 5 and then subtract 8. The answer is 27. Find the number.

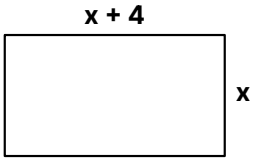
b) Three consecutive whole numbers add to 96. Find the three numbers.

c) A taxi charges \$5 to start, plus \$2 for every kilometre. A trip costs \$29. How far was the trip?

d) Tane is x years old. His mother is 4 times as old. Their ages add to 55. How old is each of them?

e) The three angles of a triangle are x°, $2x^\circ$ and $(x+30)^\circ$. Find x.

6. The rectangle has a perimeter of 44 cm. Write an equation and solve it to find x. Then find the length of the rectangle.



★ Extension FOR EARLY FINISHERS

7. Solve $4(2x-3) - 3(x+2) = 2$.

8. Solve $5x + 2 = 9 - 3x$. Give your answer as a fraction.

9. Solve $\frac{x+3}{2} + \frac{x-2}{5} = 6$.

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